Toronto Metropolitan University

Metropolitan Undergraduate Engineering Society

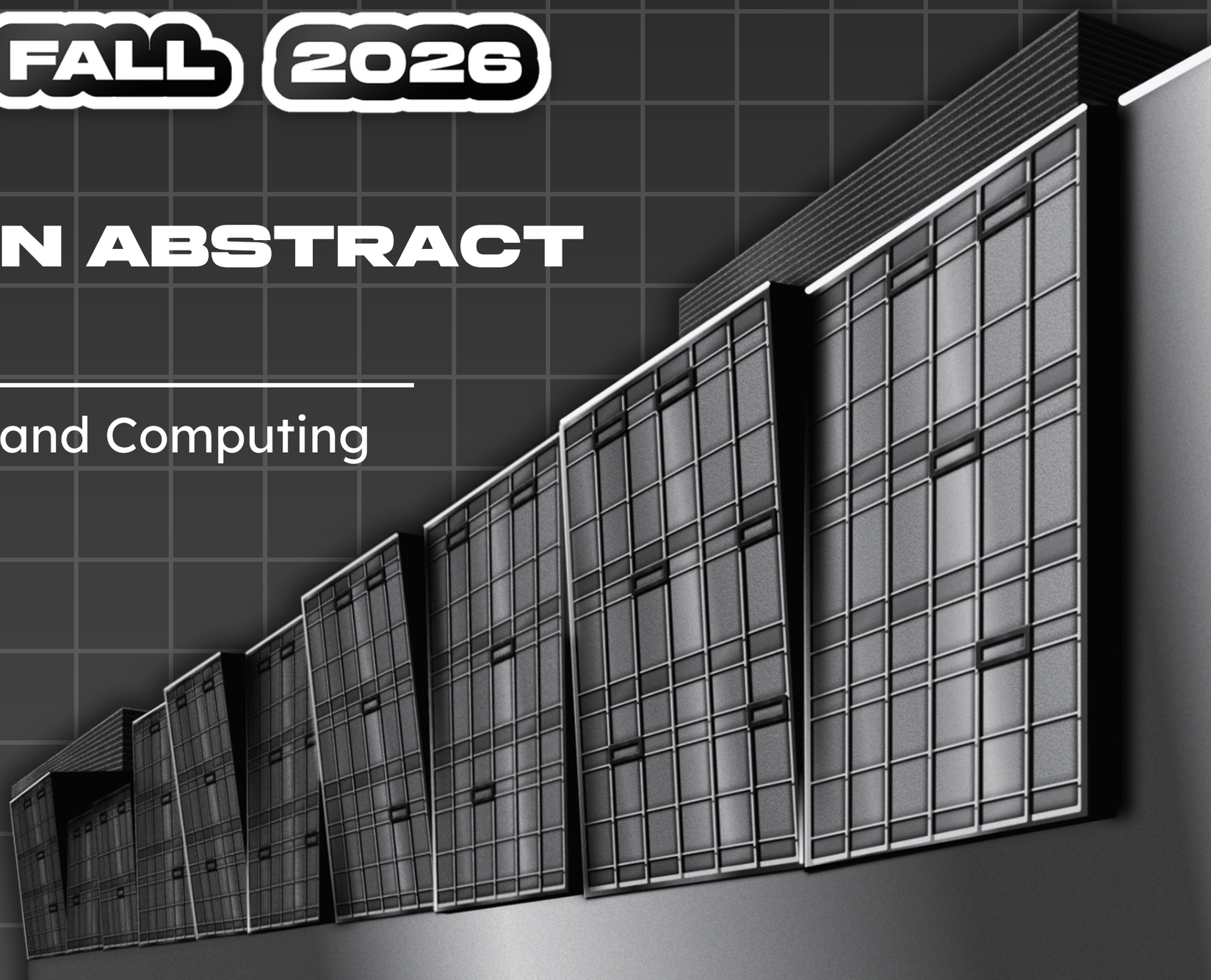
TORONTO METROPOLITAN UNIVERSITY ENGINEERING COMPETITION

FALL 2026

JUNIOR DESIGN ABSTRACT

NOVEMBER 7 & 8, 2026

George Vari Engineering and Computing Centre



METROPOLITAN ENGINEERING COMPETITION 2026
JUNIOR DESIGN
COMPETITION PACKAGE
Disaster-Resilient Insulated Food Transport Trailer
Theme: Feeding the Future: Designing Smart Infrastructure for Urban Food Security
November 7, 2026

1. INTRODUCTION

Welcome to the Metropolitan Engineering Competition (MEC) 2026! MEC provides engineering students with the opportunity to apply their technical knowledge, creativity, and problem-solving skills to practical engineering challenges.

The Junior Design Competition challenges participants to develop functional engineering solutions through hands-on prototyping, experimentation, and teamwork. Competitors must consider technical requirements, material limitations, sustainability, and practical implementation.

This year's theme, “Feeding the Future: Designing Smart Infrastructure for Urban Food Security,” emphasizes the importance of reliable and sustainable infrastructure in supporting accessible food systems.

Through this competition, participants will explore how engineering design can help communities maintain access to essential food supplies when conventional transportation systems are disrupted.

2. THE CHALLENGE

2.1 Motivation

Natural disasters such as floods, storms, and earthquakes can damage transportation infrastructure, interrupt electricity supplies, and disrupt the delivery of essential food to affected communities.

When conventional transportation routes become inaccessible, smaller and more adaptable transportation systems may be required to deliver temperature-sensitive food through narrow, uneven, or obstructed pathways.

Maintaining appropriate temperatures during transportation is especially challenging when access to refrigeration and electrical power is limited. Passive insulation, durable construction, and efficient material selection can help protect food supplies while reducing energy requirements.

The Junior Design Competition challenges participants to develop a compact transportation solution that balances thermal protection, structural integrity, maneuverability, stability, and material efficiency.

2.2 Description

Teams are required to design, construct, and demonstrate a disaster-resilient insulated food transport trailer capable of carrying a standardized ice payload through a simulated disaster-relief obstacle course.

The trailer will be towed by a standardized remote-controlled (RC) vehicle provided by the Competition Organizers. Teams are responsible for designing and constructing the trailer, including its structural frame, wheels, insulation, cargo compartment, and hitch attachment.

Competitors are not permitted to modify the RC vehicle.

The completed trailer must satisfy the following objectives:

Objective	Description
Thermal Protection	Minimize ice melting through effective passive insulation.
Obstacle Navigation	Navigate the designated obstacle course while maintaining stability and maneuverability.
Structural Integrity	Withstand course conditions without significant structural damage or component failure.
Hitch Connection	Maintain a reliable attachment to the standardized RC vehicle.
Cargo Retention	Keep the standardized ice payload contained throughout official testing.
Material Efficiency	Use approved materials and the available Engineering Dollar budget effectively.
Engineering Design	Demonstrate thoughtful decisions involving insulation, structure, weight distribution, manufacturing, and cost.
Sustainability	Consider how the design could be manufactured, maintained, and adapted for practical disaster-relief applications.

Teams will have eight (8) hours to develop their solutions, construct and test their prototypes, and prepare their required deliverables.

Each team will be permitted a maximum of three (3) supervised practice attempts during the designated testing period. Teams may request practice ice from the Competition Organizers, subject to availability.

During official testing, the trailer will carry a standardized ice payload through the obstacle course. Its performance will be evaluated based on course completion, structural integrity, cargo retention, and thermal protection.

2.3 Requirements and Constraints

Requirement	Description
Team Size	Three (3) to four (4) competitors
Working Time	Eight (8) hours
Maximum Trailer Dimensions	15 cm × 25 cm × 15 cm. All permanently attached components, including the wheels and hitch, must fit within the specified dimensional limits.
Maximum Trailer Mass	500 g, excluding the standardized ice payload. The mass includes all components permanently attached to the trailer, including its frame, wheels, axles, hitch, insulation, and cargo compartment.
Starting Budget	Each team will receive 100 Engineering Dollars (MEC$) to purchase approved materials from the MEC Item Shop.
Towing Vehicle	Standardized RC vehicle provided by MEC
Hitch Connection	The RC vehicle will have a standardized S-hook attachment provided by the Competition Organizers. Teams must design a secure trailer connection compatible with this attachment. Teams may purchase approved hitch components from the MEC Shop or construct their own using approved materials. Modifications to the RC vehicle or organizer-provided attachment are not permitted.
RC Vehicle Modification	Not permitted
Materials	Only organizer-approved materials may be incorporated into the prototype
Cargo Retention	Ice payload must remain contained during official testing

Structural Integrity	Trailer must remain stable and structurally sound
Safety	No exposed sharp edges, hazardous materials, or unsafe mechanisms
Powered Components	Trailer must remain passive unless expressly authorized
Practice Testing	Conducted without official competition ice
Official Testing	Conducted using the standardized ice payload

3. RULES

3.1 Allowed Resources

Competitors may use textbooks, course notes, and other reference materials permitted by the Competition Organizers.

Each competitor may use one (1) electronic device at a time for permitted design and documentation activities. Devices may include laptops, tablets, or similar electronic equipment.

Internet access may not be used to assist in developing the competition solution.

All materials incorporated into the completed prototype must come from the approved competition inventory. Competitors may not bring external construction materials, preassembled components, or completed prototype sections unless expressly authorized by the Competition Organizers.

3.2 Question Periods

Following the competition briefing, competitors will have an opportunity to ask questions regarding the challenge, design constraints, materials, testing procedures, and judging criteria.

During the competition, questions must be directed to the Competition Leads or through the official clarification process.

Any clarification that materially affects competition requirements will be communicated consistently to all teams.

3.3 Testing Rules

All teams will be evaluated using standardized testing conditions and procedures.

Before official testing, each trailer must undergo a standardized inspection conducted by the Competition Organizers.

The inspection will verify the trailer's dimensions, mass, approved materials, structural safety, and compatibility with the standardized RC vehicle hitch.

All permanently attached trailer components must be included during dimensional and mass measurements.

Trailers that fail to meet the mandatory requirements must be corrected within the permitted competition period before they can participate in official testing.

Each team will be permitted a maximum of three (3) practice attempts during the designated testing period, subject to coordinator supervision and equipment availability. Teams may request ice for insulation testing, subject to availability. Practice results will not contribute to the official competition score.

Each team will receive one (1) standardized ice payload with an initial mass of 50 grams for official testing. The total official testing period will be five (5) minutes, including a maximum of four (4) minutes for obstacle-course testing and the remaining time for additional thermal testing. The remaining solid ice will be weighed at the conclusion of the five-minute period to determine the percentage of ice preserved.

During official testing, the trailer will carry a standardized ice payload through the designated obstacle course.

Judges will evaluate course completion, stability, structural integrity, hitch reliability, cargo retention, and thermal performance.

Thermal performance will be calculated using:

Ice Remaining (%) = (Final Solid Ice Mass ÷ Initial Solid Ice Mass) × 100

Each team will receive a standardized ice payload with a target initial mass of 50 grams, measured using the same weighing procedure immediately before official testing. The initial and final solid ice masses will be recorded for each team, and the percentage of ice remaining will be calculated using the official competition formula.

The initial solid ice mass will be recorded before official testing. After testing, the remaining solid ice will be separated from any meltwater and weighed using the same standardized procedure.

Only the remaining solid ice mass will be used to calculate the percentage of ice preserved.

All teams will be evaluated under equivalent testing conditions. The total official testing period will be five (5) minutes, including a maximum of four (4) minutes for obstacle-course completion and the remaining time for additional thermal testing. The detailed testing procedure will be explained during the competition briefing.

During the official obstacle-course test, each team will receive one continuous timed attempt. If the trailer detaches, tips over, or cannot successfully navigate an obstacle, the Competition Organizers may allow the team to reposition the trailer at the point of failure and retry that obstacle once. If the team remains unable to complete the obstacle, the Competition Organizers may direct the team to skip it and continue along the course.

The official testing timer will not be paused or reset for retries, repositioning, or skipped obstacles. Judges will record any failures, manual interventions, or skipped obstacles and evaluate them consistently under the existing Obstacle-Course Performance criterion. No additional penalty will be imposed unless it is expressly provided in the official judging rubric.

3.4 Judging Overview

Teams will be evaluated in three categories.

Category	Maximum Points
Design Quality	40
Testing	30
Presentation	30
TOTAL	100

3.5 Testing Deductions

Violation	Points Deducted
Cargo compartment opens during testing	–5
Water leakage	–5
Ice lost due to structural failure	–10

Testing deductions will be recorded by the judges during the official testing procedure.

Each confirmed violation will be documented and applied according to the official Junior Design judging rubric.

Each testing violation will be recorded separately. Multiple deductions may apply when separate violations occur. However, the same incident must not be penalized more than once under overlapping violation categories.

3.6 Competition Penalties

Violation	Points Deducted
Plagiarism	–50
Insufficient citations	–50
Documents received after the official deadline	–50
Absent team member	–25

4. DELIVERABLES

Each team must complete the following deliverables:

Physical Prototype

A completed insulated trailer that satisfies the competition's dimensional, mass, safety, material, and hitch requirements.

Functional Demonstration

Participation in the official obstacle-course test using the standardized RC vehicle and ice payload.

Engineering Presentation

A presentation explaining the team's design process, material selection, construction methods, testing observations, engineering decisions, and design improvements. Must be a PDF.

The presentation should address:

1. Initial design concepts and sketches.
2. Material selection and budget allocation.
3. Insulation and structural design.
4. Construction and iteration.
5. Testing observations.
6. Sustainability and practical applications.
7. Final design evaluation.
8. Bill of materials and references.

4.1 Final Presentation

Each team will present its completed design to a judging panel.

Teams will have ten (10) minutes to present their solution, followed by a five (5) minute question-and-answer period.

All team members must attend and participate in the presentation.

The presentation must explain the team's design process, engineering decisions, material selection, prototype development, and testing observations.

The official presentation schedule, submission method, and submission deadline will be communicated by the Competition Organizers.

4.2 Plagiarism and Academic Integrity

All submitted designs, prototypes, and presentations must represent the team's own work completed during the designated competition period.

External sources, including technical references, images, and design inspiration, must be appropriately acknowledged.

Plagiarism, unauthorized pre-developed solutions, or deliberate misrepresentation of another person's work may result in penalties in accordance with the official MEC competition rules.

5. MATERIALS

5.1 Materials Provided

The Competition Organizers will provide the following tools and materials to each Junior Design team at no Engineering Dollar cost.

Tool / Material	Quantity Per Team	Return or Keep
Mini Hot Glue Gun	1	Return
Scissors	1	Return
X-Acto / Utility Knife	1	Return
Plastic Ruler	1	Return
Mini Hot Glue Sticks	5	Keep
HB #2 Pencil	1	Keep
Gardening Gloves	1 pair	Return
Long-Nose Pliers	1	Return
7 in. Coping Saw with Blades	1	Return
11-Piece Hand Drill Set	1 set	Return

Screwdriver Bit	1	Return
1/4 in. Screwdriver Handle	1	Return
Disposable Masks	2	Keep
Coloured Electrical Tape	1 roll	Keep
Eraser	1	Keep

5.2 MEC Shop Items

Each Junior Design team will receive 100 Engineering Dollars (MEC$) to purchase approved materials from the MEC Item Shop.

All materials available through the MEC Item Shop have assigned Engineering Dollar prices and may be subject to purchase limits and inventory restrictions.

Teams must remain within their allocated Engineering Dollar budget. Unused Engineering Dollars will not be exchanged for additional competition points or other benefits.

Materials are available in limited quantities. Teams are encouraged to plan their purchases carefully and consider alternative materials if certain items become unavailable.

Ordering procedures, purchase restrictions, and applicable refund or exchange policies will be communicated by the Competition Organizers.

6. APPENDIX

6.1 MEC Shop Price List

The following materials are available for purchase through the MEC Item Shop. All prices are listed in Engineering Dollars (MEC$). Items are subject to availability and the maximum purchase limits indicated below.

Item	Price (MEC$)	Maximum Per Team
1/4 in. Poly Rope - 3 ft Portion	5	3
Balsa Wood Sheet - 12 × 12 × 1/8 in.	8	3

Polyester Batting - 50 g Portion	6	2
Rubber Weather Stripping - 1 ft Portion	6	2
Foam Wheel	15	2
Stainless Steel Latch	14	2
Small Compression Springs - 2 Piece Pack	4	10
Hook-and-Loop Fastener - 10 cm	3	10
Wooden Dowels - 2 Pieces	5	5
Rubber Caster Wheel	15	2
M3 Bolt	1	30
M3 Nut	1	30
M3 Flat Washer	1	20
626ZZ Ball Bearing - 6 mm ID × 19 mm OD	10	2
5 mm Stainless Steel Pad Eye Hook	2	2
Mini Metal Hinges - 2 Set Pack	4	3
Clear Acrylic Sheet - 100 × 100 × 1 mm	16	1
Bubble Wrap - 1 ft × 1 ft	6	3
Clear Vinyl Tubing - 1 ft Portion	4	3
Double Reflective Insulation - 24 × 12 in.	15	2
Large Hook-and-Loop Tape - Prepared Portion	10	1
Rubber Bands - 5 Piece Pack	4	3
Thick White Twine String - 10 ft	6	2
Stainless Steel Flat Washers - 10 Piece Pack	4	2

Zip Ties - 2 Piece Pack	4	2
Large Binder Clip	4	4
Small Binder Clips - 2 Piece Pack	3	2
Wooden Clothespins - 4 Piece Pack	4	2
Silicon Carbide Sandpaper - 1 Sheet	4	3
Regular Popsicle Sticks - 10 Piece Pack	5	2
Bamboo Skewers - 10 Piece Pack	5	2
Coloured Paper Straws - 3 Piece Pack	3	2
White Foam Board - 20 × 30 in.	20	1
Corrugated Cardboard Sheet - 8.5 × 11 in.	6	1
Mini Hot Glue Sticks - 2 Piece Pack	6	2
Compression / Extension Spring - 1 Piece	3	8

6.2 Standardized Testing Specifications

The following specifications establish the requirements for the official Junior Design testing procedure. All teams will be evaluated under equivalent conditions using the standardized equipment and procedures provided by the Competition Organizers.

Component	Specification
RC Vehicle	Standardized MEC-provided 1:18 brushless 4WD RC car
RC Vehicle Quantity	One organizer-controlled RC vehicle will be used for official testing
Hitch Interface	Standardized attachment point provided by the Competition Organizers
Ice Payload	One standardized ice payload with an initial mass of 50 grams

Official Test Duration	Five (5) minutes total, including a maximum of four (4) minutes for obstacle-course testing and the remaining time for additional thermal testing.
Practice Attempts	Maximum of three (3) supervised practice attempts per team
Official Attempts	One (1) official scored attempt per team
Maximum Trailer Dimensions	15 cm × 25 cm × 15 cm, including permanently attached components
Maximum Trailer Mass	500 g, excluding the standardized ice payload
RC Vehicle Modification	Competitors are not permitted to modify the standardized RC vehicle.
Ice Measurement	Percentage of solid ice remaining after official testing
Obstacle Course	Standardized course layout established by the Competition Organizers